

Accessible Emotion-Aware E Learning Platform for Deaf Students Studying O/L ICT Using Facial Expression Recognition and an NLP Based Chatbot

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INTRODUCTION

The Problem and Stakeholders Impact on Users & Society

Identifying challenges and impacted users in digital education

The Problem

01

1 Digital learning platforms are widely used in education
2 Most systems rely on audio explanations and large text content
3 Deaf students struggle to understand learning materials

Users & Stakeholders

02

1 Digital Deaf and hard-of-hearing O/L IT students
2 Teachers and special education instructors
3 Schools and educational institutions

- Learning confusion and frustration
- Reduced engagement in technical subjects
- Limited access to equal education opportunities

INTRODUCTION

Existing Solutions and Limitations

Analyzing current educational technology gaps

Existing Solutions

1. E-learning platforms: Moodle Coursera and Khan Academy
2. Provide captions and subtitles

01

Limitations

1. Content is static and not adaptive
2. Cannot detect student emotional states
3. Chatbots lack emotion-aware learning support

02



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INTRODUCTION

Proposed Solution and Contributions

A multi-faceted approach to inclusive e-learning

Proposed Solution

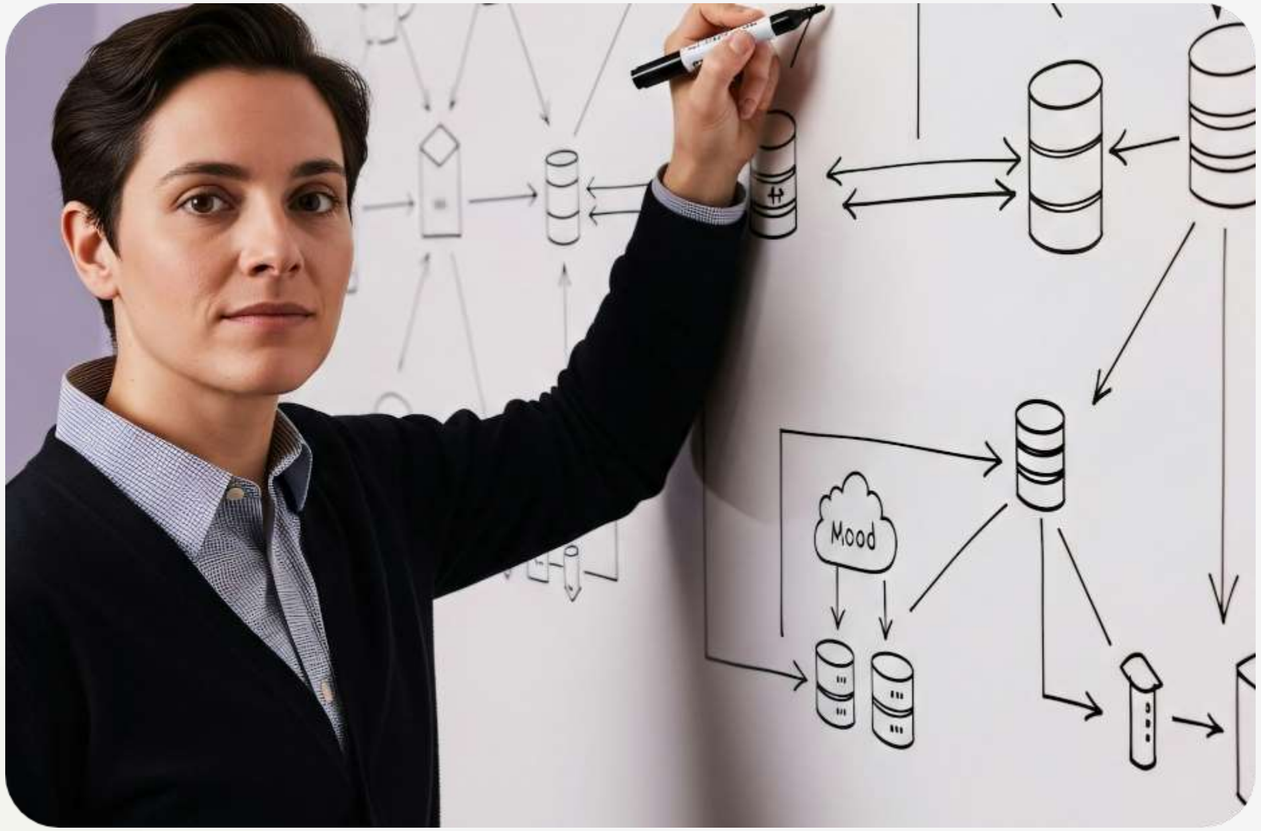
01

1. Real-time facial emotion detection
2. Adaptive visual learning content
3. NLP-based intelligent chatbot

Individual Contributions

02

1. Emotion Detection Subsystem
2. Adaptive Learning Content System
3. Secure Communication Framework
4. NLP-Based Chatbot Learning Assistant





IT22211200

Real Time Emotion Detection Subsystem



IT22211200 Tharaka W A K N R26-IT-18

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EMOTION DETECTION

Subsystem Core Overview

Contribution Problem and Novelty of Emotion Detection



Contribution: Implement real-time facial expression recognition FER to detect learner emotions such as engagement confusion and frustration and transmit emotional data to the adaptive learning module and NLP chatbot



Problem: No existing system provides emotion-aware deaf-tailored real-time emotion detection subsystem for O/L IT learning.



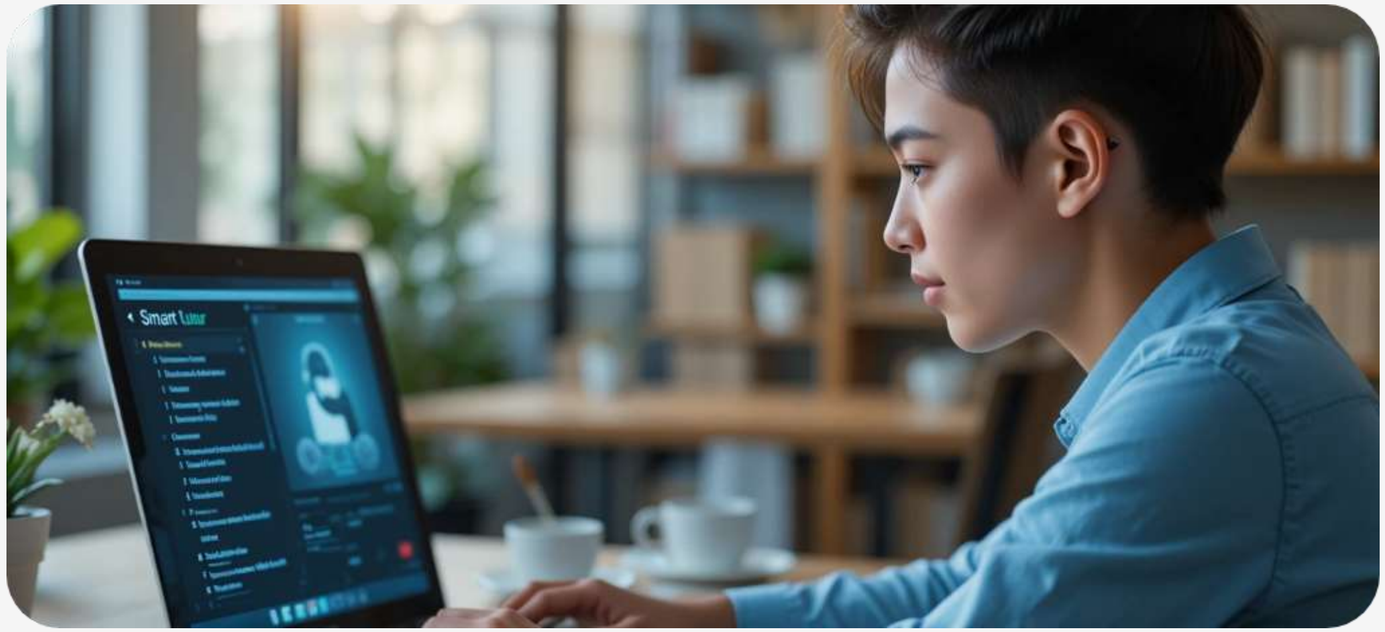
Novelty: Real-time emotion detection tailored for deaf learners that enables dynamic adaptation of visual and text-based learning content a capability absent in current e-learning systems.

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EMOTION DETECTION

Domain Knowledge and Methodology

Technologies and research approach for emotion recognition



Objective :

Develop a real-time facial emotion detection subsystem to identify learner emotional states and support adaptive learning for deaf O/L IT students

Research Methodology

Design Science Research DSR approach

Real-time facial emotion detection system development

Deep learning model training and testing for emotion classification

Knowledge Domain:

Computer Vision and Machine Learning for facial expression recognition

Technologies Used:

- OpenCV for face detection
- TensorFlow/PyTorch for CNN-based emotion classification
- FER-2013 and CK+ datasets for model training

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EMOTION DETECTION

Validation and Data Considerations

Ensuring accuracy and ethical standards



Validation & Evaluation:

- Emotion classification evaluated using accuracy precision recall and F1-score
- Real-time performance measured using FPS benchmarks
- Validation using facial expression data from deaf learners

01

Data Security & Ethical Clearance:

FER-2013 and CK+ public datasets

Controlled experiments with voluntary deaf participants

Anonymized facial data handled under ethical research approval

02

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EMOTION DETECTION

System Overview

Dependencies and common components of the detection system

- Webcam input for capturing facial expressions
- OpenCV for face detection and preprocessing
- TensorFlow / PyTorch CNN model for emotion classification
- Emotion API / Backend server for transmitting emotion data
- Integration with Adaptive Learning Module and NLP Chatbot

PHASE 1
Webcam Input Student Face

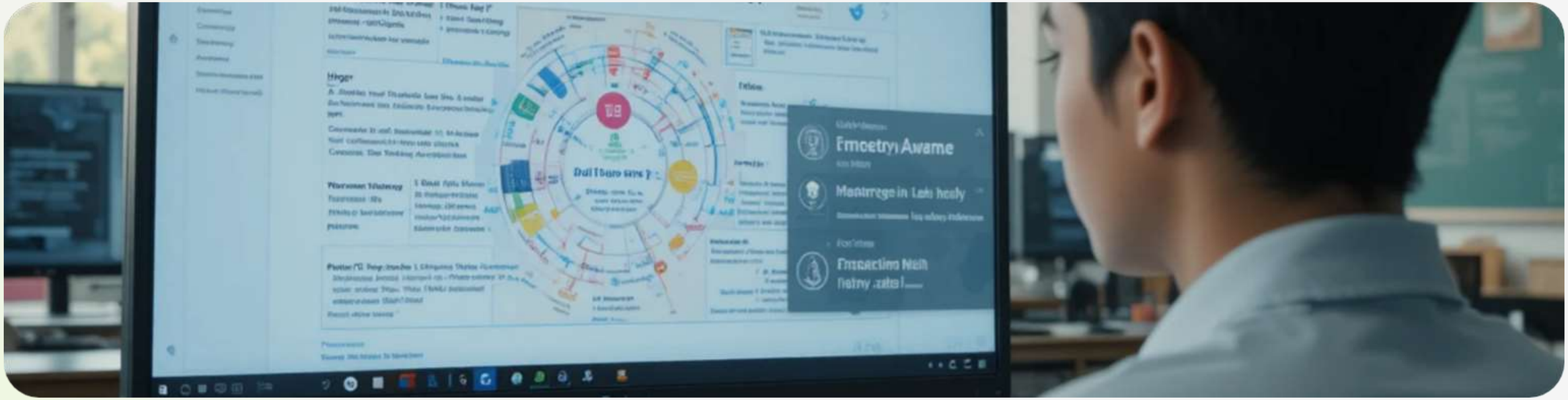
PHASE 2
Face Detection Open CV

PHASE 3
Image Preprocessing & Feature Extraction

PHASE 4
CNN Emotion Classification Model TensorFlow/ PayTorch

PHASE 5
Detect Emotion & Emotion API Backend Server JSON Data Transmission

PHASE 6
Aduptive Learning Modul & NLP Chat Bot Adjust Learning Content



IT2219776

Adaptive Learning Content Delivery System



IT2219776 Mendis L B L R26-IT-18

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ADAPTIVE LEARNING

SubSystem Core Overview

Contribution Novelty and Problem Identification

Contribution:

Develop an adaptive learning content engine that dynamically adjusts the complexity and visual presentation of O/L IT learning materials based on real-time emotional and engagement data from learners.

Problem:

Traditional e-learning systems deliver static learning content and cannot dynamically adapt explanations or visual materials when deaf students experience confusion or frustration during complex IT learning tasks.

Novelty:

An emotion-aware adaptive learning engine that proactively modifies visual and textual learning content in real time based on detected learner emotional states a capability largely absent in existing elearning systems designed for deaf students.



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ADAPTIVE LEARNING

Domain Knowledge and Methodology

Technologies and research approach for emotion recognition



Research Objective

1 Develop an adaptive learning engine that modifies content based on learner emotional state

Research Methodology

- Design Science Research DSR approach
- Development of an emotion-aware adaptive learning engine
- Rule-based algorithm for dynamic content adaptation

Key Knowledge Domains

- Artificial Intelligence and Adaptive Learning Systems
- Affective Computing
- Human–Computer Interaction HCI

Technologies Used

- Python for backend adaptive learning logic
- React.js for dynamic user interface adaptation
- Mongo DB & CDN for hierarchical learning content storage
- Integration with emotion recognition and interaction tracking modules

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ADAPTIVE LEARNING

Validation and Data Considerations

Ensuring accessibility and privacy in content delivery



01

Validation & Evaluation:

- Pre/post learning outcome assessments
- Engagement metrics
- WCAG 2.1 accessibility audit.

02

Data Security & Ethical Clearance:

- Emotion and learner interaction data collected for analysis.
- Learning datasets based on the O/L IT curriculum.
- User consent required for webcam and interaction data.
- Data stored anonymously with encrypted communication.

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ADAPTIVE LEARNINGEMOTION DETECTION

System Overview

Dependencies and common components of the detection system

- Emotion Detection Subsystem for emotional state data
- Interaction Tracking Module for learner engagement monitoring
- Adaptive Content Engine for rule-based content adaptation
- React.js Frontend for dynamic UI updates
- Integration with learning content repository

PHASE 1

Emotion Recognition Module Detect Emotional State

PHASE 2

Interaction Tracking Module Mouse Scroll Engagement

PHASE 3

Aduptive Content Engine Decision Algorithem / Rule Logic

PHASE 4

Content Selection Text / Simplified / Visual Content

PHASE 5

React.js Frontend Dynamic UI Update

PHASE 6

Adapted Learning Content Improved Understanding

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NLP Chatbot Based Intelligent Subsystem

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INTELLIGENT CHATBOT

SubSystem Core Overview

Contribution Problem and Novelty of the NLP Assistant

Contribution:

01

Develop an NLP-based intelligent tutoring chatbot that interprets student queries and delivers simplified curriculum aligned explanations with visual learning support based on real-time emotional feedback.

Problem:

02

Traditional e-learning platforms provide static learning materials and lack conversational tutoring support that can simplify complex IT concepts and respond adaptively when deaf students experience confusion during learning.

Novelty:

03

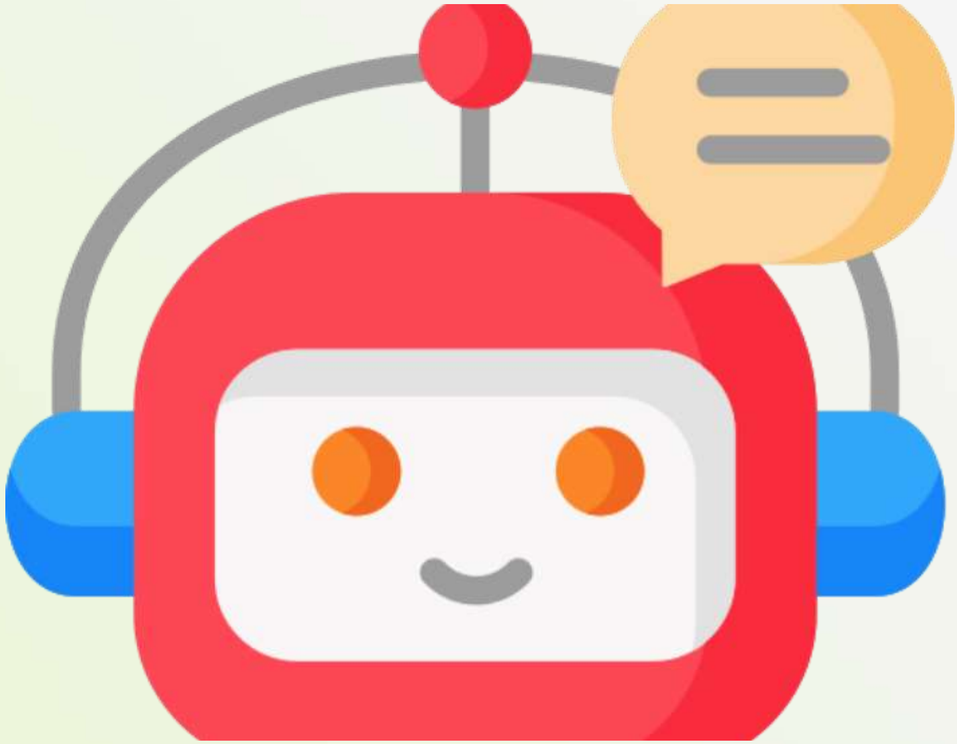
An emotion-aware NLP tutoring chatbot that dynamically adapts explanations and triggers visual learning aids based on both student queries and real-time emotional states a capability largely absent in existing e-learning systems.

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INTELLIGENT CHATBOT

Domain Knowledge and Methodology

Conversational AI and Deep Learning frameworks



Research Objectives:

Develop an intelligent NLP-based chatbot subsystem that provides adaptive tutoring support for deaf students studying the G.C.E. O/L Information Technology syllabus.

01

Research Methodology:

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- Design Science Research DSR approach
- Development of an NLP-based intelligent tutoring chatbot
- Integration of NLP emotion data and adaptive learning content

02

Knowledge Domains:

- Natural Language Processing NLP
- Artificial Intelligence and Conversational AI
- Human–Computer Interaction HCI

03

Technologies Used

- Python for chatbot backend logic Hugging Face Transformers / spaCy / NLTK for NLP processing
- TensorFlow / PyTorch for deep learning models
- React.js for chatbot interface and interaction MongoDB for storing chatbot conversation

04

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INTELLIGENT CHATBOT

Validation and Data Considerations

Measuring accuracy and user satisfaction



Validation & Evaluation:

- Intent recognition accuracy of the NLP model
- Response quality using BLEU / ROUGE scores
- Pre-test / Post-test to measure learning improvement SUS survey to evaluate usability and user satisfaction

Data Security & Ethical Clearance:

O/L IT curriculum materials and chatbot interaction logs- Emotion metadata from the emotion detection subsystem- Informed user consent and anonymized learner data- Secure communication using HTTPS/TLS encryption

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INTELLIGENT CHATBOT

System Overview

Dependencies and common components of the detection system

- Emotion Detection Subsystem for learner emotional state data
- Adaptive Learning Content Subsystem for visual learning materials
- NLP processing module for query understanding
- Backend API and database MongoDB for interaction management
- React.js chatbot interface for user interaction

PHASE 1

Student Chat Input

PHASE 2

NLP Processing Intent Detection

PHASE 3

Emotion Metadata Integration

PHASE 4

Chatbot Decision Engine

PHASE 5

Aduptive Learning Content Visual Aids / Diagrams

PHASE 6

Chatbot Response





IT19131948

Secure and Optimized Data Communication Framework

IT19131948 Nawarathna I K K R B R26-IT-18

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COMMUNICATION FRAMEWORK

Communication Core Overview

Contribution Problem and Novelty of the Framework

Contribution:

Develop a secure and bandwidth-optimized communication framework that ensures protected efficient and low-latency transmission of learner-related data between the client browser and backend server of the emotion-aware elearning platform.

Research Contribution

Many web-based educational systems prioritize application functionality while giving limited attention to communication security and bandwidth efficiency leading to increased latency higher bandwidth usage and potential exposure of sensitive learner related data.

Novelty :

A secure and bandwidth optimized communication framework that integrates HTTPS/TLS encryption payload minimization and HTTP compression to improve transmission efficiency and protect sensitive learner-related data in real-time educational web applications.

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COMMUNICATION FRAMEWORK

Network Domain and Methodology

Web performance and cryptographic optimization

1

Research Objectives:

Design and evaluate a secure and bandwidth-efficient communication framework that improves network performance and protects learner data in the emotion-aware e-learning platform.

2

Research Methodology:

- Design Science Research (DSR) approach
- Design of secure client–server communication framework
- Evaluation of security and bandwidth optimization techniques

3

Knowledge Domain:

- Computer Networks and Web Communication
- Network Security and Cryptography
- Web Performance Optimization

4

Technologies Used:

- HTTPS / TLS encryption for secure client–server communication
- HTTP compression GZIP / Brotli for bandwidth optimization
- REST API architecture for efficient data exchange
- Node.js / Express backend and React.js frontend communication

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COMMUNICATION FRAMEWORK

Validation and Data Considerations

01

Validation & Evaluation:

Measure network latency and response time- Evaluate bandwidth usage and payload size reduction- Compare performance with non-optimized communication models- Analyze security overhead introduced by encryption

02

Data Security & Ethical Clearance:

Learner interaction data and system communication logs- Emotion metadata exchanged between system modules- Secure transmission using HTTPS/TLS encryption- User consent and anonymized data for privacy protection

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COMMUNICATION FRAMEWORK

System Overview

Dependencies and common components of the detection system

- Client Web Browser Interface
- Secure Communication Layer HTTPS/TLS
- Backend Application Server / API services
- Database for learner data and system logs
- Compression and payload optimization mechanisms

PHASE 1

Student Web Browser

PHASE 2

HTTPS / TLS Encription Secure Data Transmission

PHASE 3

Backend Application Server API Processing

PHASE 4

Payload ptimaization Data Minimizing & HTTP Compression GZIP/ Brotli

PHASE 5

Database Learner Data & Logs

PHASE 6

Compress Secure Respons to Student Web Browser

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ACCESSIBLE EMOTION-AWARE E-LEARNING PLATFORM

Business and Impact Strategy

Market Investment and Value Proposition



Investment & Cost Recovery:

- Labour 4 students 6 months
- Cloud hosting & dataset costs
- Revenue: Institutional licensing + SaaS subscription

01

Target Market:

1. Deaf O/L IT students in Sri Lanka
2. Schools for the Deaf
3. Ministry of Education EdTech initiatives
4. International special education platforms

02

Value Proposition

1. First emotion-aware adaptive EdTech platform for deaf IT learners
2. Improves engagement comprehension & exam outcomes

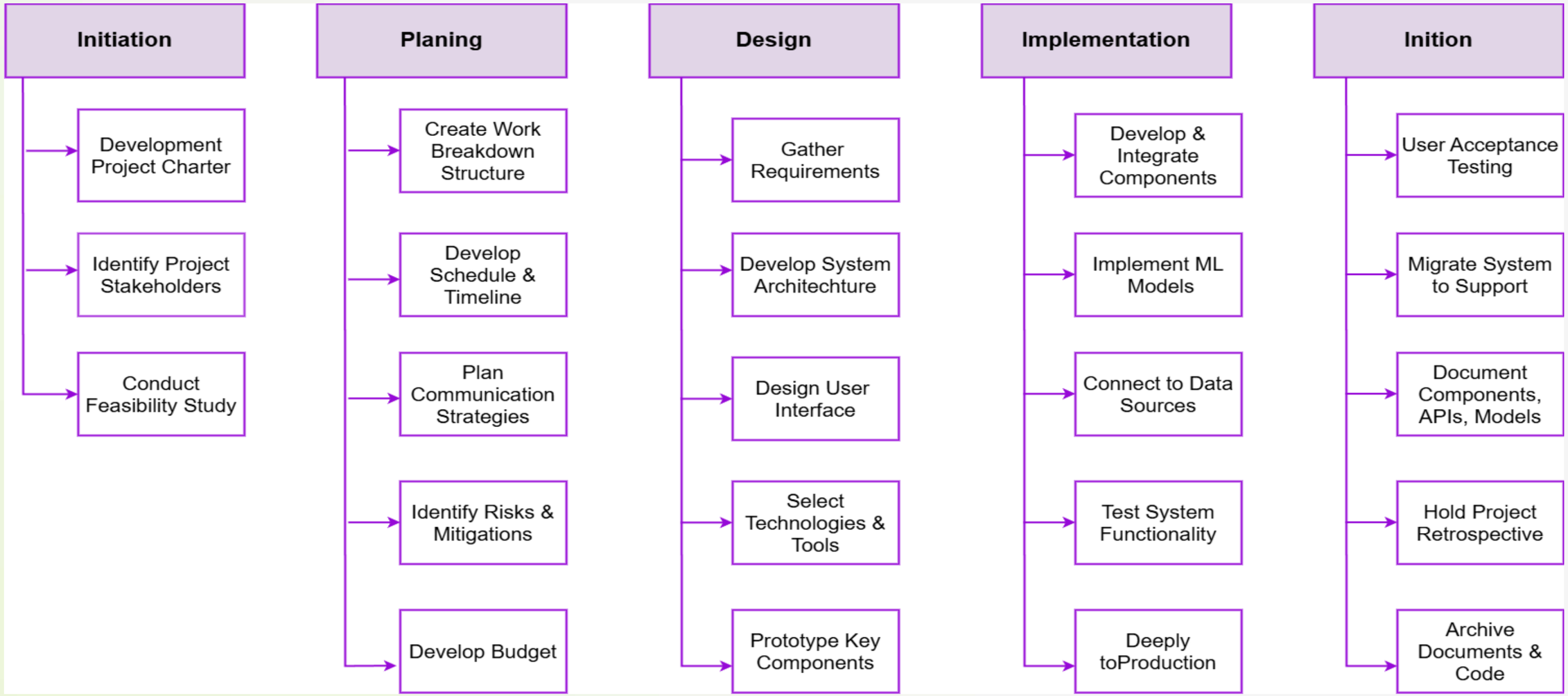
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Intellectual Property

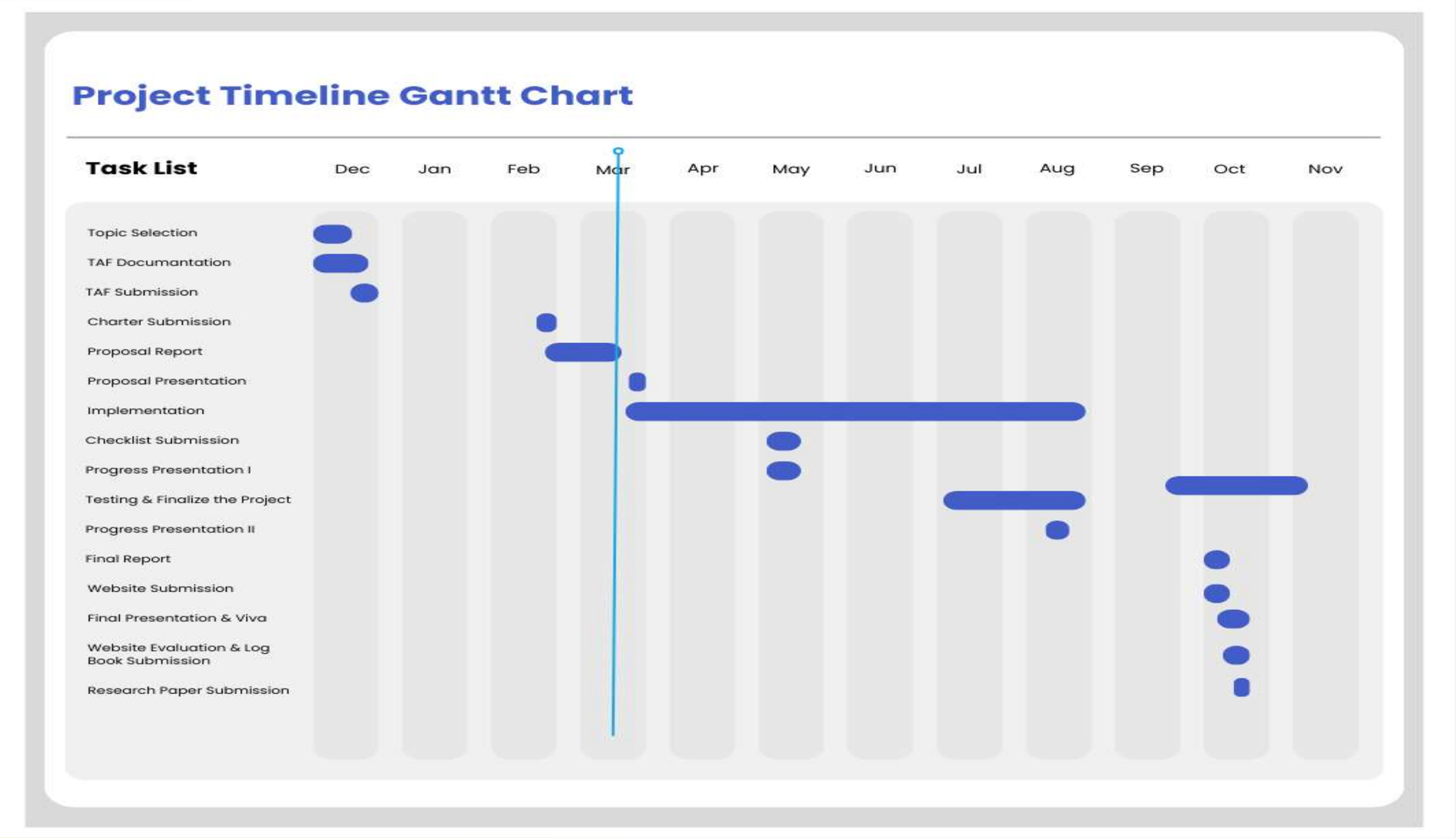
1. System architecture & adaptive algorithms
2. NLP chatbot framework
3. Supports SDG-4 & SDG-10

04

Accessible Emotion-Aware E-Learning Platform



Accessible Emotion-Aware E-Learning Platform



THANK YOU SO MUCH

We appreciate your questions and feedback — please share any



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